AGML Lab: Week 6
Decision Trees

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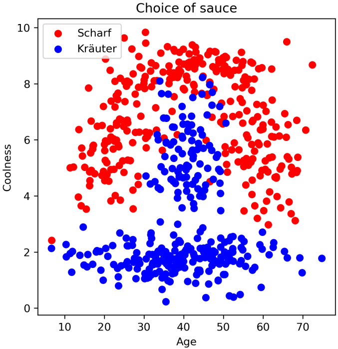
03.06.2026

(Jupyter Notebook)

Problem Statement

What sauce do you want on your kebab?

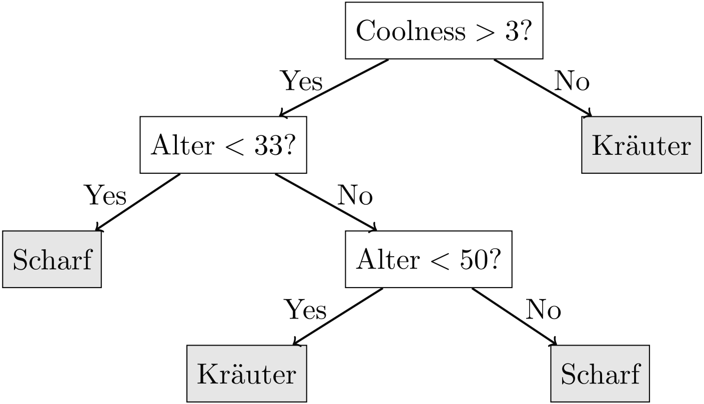


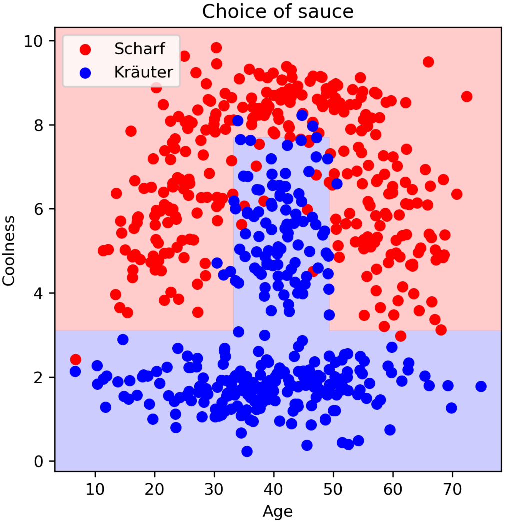
Looks like an easy distribution, so i want a classifier i can use in my head.

Decision Trees:

- easy to interpret
- works on small datasets
- robust to outliers

Idea: use a tree of questions to classify the data



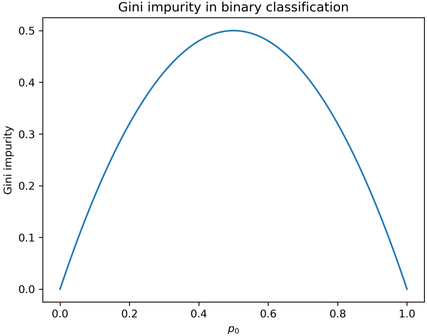


Choice of sauce
Scharf
Kräuter
Coolness
Age
0
2
4
6
8
10
10
20
30
40
50
60
70

How to learn a decision tree? $\rightarrow$ we need good questions
How to learn a good question? $\rightarrow$ measure of how mixed your dataset is after the question: Gini Impurity

$$p_c(y) = \frac{\#\ c \text{ in } y}{\#\ y}$$

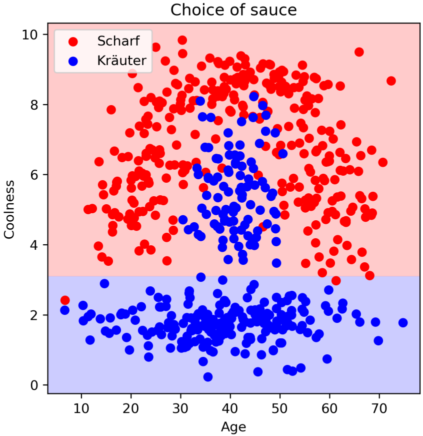
$$\text{Gini}(y) = 1 - \sum_{c=1}^{k} p_c(y)^2$$



Now we just combine the impurity of the regions:

$$S(y_0, y_1) = \frac{|y_0|}{|y_0| + |y_1|} \cdot \mathrm{Gini}(y_0) + \frac{|y_1|}{|y_0| + |y_1|} \cdot \mathrm{Gini}(y_1)$$

and greedily minimize $S(y_0, y_1)$ for every question.



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Exercises

1. have a nice day
2. get enough sleep